

Sealed Experiment Record

Sealed 2026-10-08 · Signed by the K-Veritas server
Session 4c96ebf6 · Data hash 902e...c091

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-08 08:47 and 08:51 UTC. Reported results: first_loss 7.98317, final_loss_mean_last100 1.7264. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	4c96ebf6-5a3c-47ab-8cfb-6e08fd8770b7
Initialized	2026-10-08T08:47:51Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	8e6439b034a4f3601322064c0dde63aa2a2eb2ab7e91450b6c849dca832083f2
Disclosure	open
Source code, run 1	cc63d0afa4e5fd3b38a308e478de62f306118bf01f4a881cc695ac57cb8f322b

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	3 minutes 11 seconds	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
loss	1.68376	2000 (62 readings)	explicit
grad_norm	0.4213	(21 readings)	heuristic
tflops	27.28	(21 readings)	heuristic
mfu	8.74	(21 readings)	heuristic

Metric	Final value	Step	Source
mean_tokens_per_second	204465		explicit

Inline claim	Value	Phase	Output line
first_loss	7.98317		65
final_loss_mean_last100	1.7264		66

4 Execution evidence

Coherence

Coherent (0.46): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	1846	205.4 s	12.50 GB	70%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 773 tracked files, 3 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	08:47:52	54fbc5fd282e4584	773
2	run_end	08:51:03	54fbc5fd282e4584	0

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 774 files read, 1 files written, 25 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-08T08:51:05.194553+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	902ef28d56b69e5748e88bcc531d53db9a87875582631ae2eb9a8a81aae3c091
Code bundle hash	9f8a20e45d71db5db150c421d2d32eccf42183e902e66d000b0e76348ec457a7

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
loss	7.98317		2	heuristic
grad_norm	1.3662		2	heuristic
tflops	0.27		2	heuristic
mfu	0.09		2	heuristic
loss	2.75805		3	heuristic
grad_norm	0.5993		3	heuristic
tflops	10.35		3	heuristic
mfu	3.32		3	heuristic
loss	2.54121		4	heuristic
grad_norm	0.7559		4	heuristic
tflops	12.27		4	heuristic
mfu	3.93		4	heuristic
loss	2.38243		5	heuristic
grad_norm	0.4382		5	heuristic
tflops	15.21		5	heuristic
mfu	4.87		5	heuristic
loss	2.52854		6	heuristic
grad_norm	0.5715		6	heuristic
tflops	15.82		6	heuristic
mfu	5.07		6	heuristic
loss	2.27117		7	heuristic
grad_norm	0.6697		7	heuristic
tflops	25.05		7	heuristic
mfu	8.03		7	heuristic
loss	2.19493		8	heuristic
grad_norm	0.5769		8	heuristic
tflops	13.83		8	heuristic
mfu	4.43		8	heuristic
loss	2.20761		9	heuristic
grad_norm	0.4164		9	heuristic
tflops	14.28		9	heuristic
mfu	4.58		9	heuristic
loss	1.99662		10	heuristic
grad_norm	0.5877		10	heuristic
tflops	19.23		10	heuristic
mfu	6.16		10	heuristic
loss	1.99469		11	heuristic
grad_norm	0.5427		11	heuristic
tflops	20.23		11	heuristic
mfu	6.48		11	heuristic
loss	1.99049		12	heuristic
grad_norm	0.483		12	heuristic
tflops	18.29		12	heuristic
mfu	5.86		12	heuristic

Metric	Value	Step	Line	Source
loss	1.99752		13	heuristic
grad_norm	0.6423		13	heuristic
tflops	11.01		13	heuristic
mfu	3.53		13	heuristic
loss	1.76484		14	heuristic
grad_norm	0.5958		14	heuristic
tflops	10.52		14	heuristic
mfu	3.37		14	heuristic
loss	2.03515		15	heuristic
grad_norm	0.4173		15	heuristic
tflops	15.02		15	heuristic
mfu	4.82		15	heuristic
loss	2.12643		16	heuristic
grad_norm	0.6245		16	heuristic
tflops	13.76		16	heuristic
mfu	4.41		16	heuristic
loss	1.80222		17	heuristic
grad_norm	0.5758		17	heuristic
tflops	16.68		17	heuristic
mfu	5.35		17	heuristic
loss	1.87624		18	heuristic
grad_norm	0.4876		18	heuristic
tflops	16.39		18	heuristic
mfu	5.25		18	heuristic
loss	1.89977		19	heuristic
grad_norm	0.4895		19	heuristic
tflops	9.99		19	heuristic
mfu	3.2		19	heuristic
loss	1.73055		20	heuristic
grad_norm	0.4358		20	heuristic
tflops	10.53		20	heuristic
mfu	3.38		20	heuristic
loss	1.7835		21	heuristic
grad_norm	0.5217		21	heuristic
tflops	12.95		21	heuristic
mfu	4.15		21	heuristic
loss	1.68376		22	heuristic
grad_norm	0.4213		22	heuristic
tflops	27.28		22	heuristic
mfu	8.74		22	heuristic
loss	7.98317	1	23	explicit
loss	2.90646	50	24	explicit
loss	2.75805	100	25	explicit
loss	2.69824	150	26	explicit
loss	2.54121	200	27	explicit
loss	2.54404	250	28	explicit
loss	2.38243	300	29	explicit
loss	2.47799	350	30	explicit
loss	2.52854	400	31	explicit
loss	2.2436	450	32	explicit

Metric	Value	Step	Line	Source
loss	2.27117	500	33	explicit
loss	2.25746	550	34	explicit
loss	2.19493	600	35	explicit
loss	2.11355	650	36	explicit
loss	2.20761	700	37	explicit
loss	1.89493	750	38	explicit
loss	1.99662	800	39	explicit
loss	2.20612	850	40	explicit
loss	1.99469	900	41	explicit
loss	2.03445	950	42	explicit
loss	1.99049	1000	43	explicit
loss	1.92345	1050	44	explicit
loss	1.99752	1100	45	explicit
loss	1.87593	1150	46	explicit
loss	1.76484	1200	47	explicit
loss	1.91363	1250	48	explicit
loss	2.03515	1300	49	explicit
loss	1.86578	1350	50	explicit
loss	2.12643	1400	51	explicit
loss	1.78788	1450	52	explicit
loss	1.80222	1500	53	explicit
loss	1.8746	1550	54	explicit
loss	1.87624	1600	55	explicit
loss	1.77202	1650	56	explicit
loss	1.89977	1700	57	explicit
loss	1.91521	1750	58	explicit
loss	1.73055	1800	59	explicit
loss	1.74193	1850	60	explicit
loss	1.7835	1900	61	explicit
loss	1.59993	1950	62	explicit
loss	1.68376	2000	63	explicit
mean_tokens_per_second	204465		64	explicit

Phase	Line	Time	CPU time	Memory	GPU
train	1	08:47:52	1.6 s	0.65 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	08:47:52	54fbc5fd282e4584	773
1	phase train	08:47:52	54fbc5fd282e4584	0
2	run_end	08:51:03	54fbc5fd282e4584	0

Event	Path or command	Hash
read	kv_run.sh	
read	run_train.sh	
read	.ci/docker/build.sh	
read	.ci/docker/common/install_base.sh	

Event	Path or command	Hash
read	.ci/docker/common/install_clang.sh	
read	.ci/docker/common/install_conda.sh	
read	.ci/docker/common/install_deepest.sh	
read	.ci/docker/common/install_gcc.sh	
read	.ci/docker/common/install_user.sh	
read	.ci/docker/common/utls.sh	
read	.claude/skills/cuda_graph_trace_compaction/scripts/compact_cuda_graph_tra...	
read	.claude/skills/cuda_graph_trace_compaction/scripts/test_compact_cuda_grap...	
read	.claude/skills/numerics_debugging/scripts/activation_tracer.py	
read	.claude/skills/numerics_debugging/scripts/compare_numerics.py	
read	.github/ISSUE_TEMPLATE/bug.yml	
read	.github/ISSUE_TEMPLATE/config.yml	
read	.github/dependabot.yml	
read	.github/labeler.yml	
read	.github/pytorch-probot.yml	
read	.github/scripts/install_deepest_v2.sh	
read	.github/scripts/integration_test_real_pg.sh	
read	.github/scripts/run_8xgpu_integration_tests.sh	
read	.github/scripts/todo_debt_tracker.py	
read	.github/scripts/update_version.sh	
read	.github/scripts/validate_release_gpu.sh	
read	.github/workflows/build_whl_and_publish.yaml	
read	.github/workflows/claude-code.yml	
read	.github/workflows/docker-builds.yml	
read	.github/workflows/integration_test.yaml	
read	.github/workflows/integration_test_8gpu_graph_trainer.yaml	
read	.github/workflows/integration_test_8gpu_graph_trainer_h100.yaml	
read	.github/workflows/integration_test_8gpu_rl.yaml	
read	.github/workflows/integration_test_8gpu_torchft.yaml	
read	.github/workflows/integration_test_8gpu_transformers_modeling_backend.ya...	
read	.github/workflows/integration_test_b200.yaml	
read	.github/workflows/integration_test_b200_graph_trainer.yaml	
read	.github/workflows/integration_test_h100.yaml	
read	.github/workflows/integration_test_rl_b200.yaml	
read	.github/workflows/integration_test_rl_h100.yaml	
read	.github/workflows/integration_test_rocm.yaml	
read	.github/workflows/lint.yaml	
read	.github/workflows/release.yml	
read	.github/workflows/set-matrix.yaml	
read	.github/workflows/test_release.yml	
read	.github/workflows/todo_debt_tracker.yaml	
read	.github/workflows/unit_test_cpu.yaml	
read	.github/workflows/unit_test_cpu_rl.yaml	
read	.github/workflows/unit_test_cpu_torchft.yaml	
read	.github/workflows/unit_test_gpu.yaml	
read	.github/workflows/unit_test_gpu_rl.yaml	
read	.github/workflows/validate_rc.yaml	
read	.github/workflows/validate_release_gpu.yml	
read	.pre-commit-config.yaml	
read	kv_recipes.py	

Event	Path or command	Hash
read	pyproject.toml	
read	scripts/_checkpoint_test_config.py	
read	scripts/checkpoint_compat_test.py	
read	scripts/checkpoint_conversion/convert_ema_to_hf.py	
read	scripts/checkpoint_conversion/convert_from_hf.py	
read	scripts/checkpoint_conversion/convert_to_hf.py	
read	scripts/checkpoint_conversion/numerical_tests_example.py	
read	scripts/checkpoint_conversion/numerical_tests_kimi.py	
read	scripts/checkpoint_conversion/numerical_tests_kimi_k3.py	
read	scripts/checkpoint_conversion/numerical_tests_qwen3_5.py	
read	scripts/checkpoint_conversion/numerical_tests_qwen3_5_shard.py	
read	scripts/checkpoint_conversion/utils.py	
read	scripts/ci/pytorch_ci_test_runner.sh	
read	scripts/download_hf_assets.py	
read	scripts/loss_compare.py	
read	scripts/rl/loss_compare.py	
read	tests/__init__.py	
read	tests/assets/c4_test/data.json	
read	tests/assets/cc12m_test/pack_test_dataset.py	
read	tests/assets/flux_test_encoders/clip-vit-large-patch14/config.json	
read	tests/assets/flux_test_encoders/t5-v1_1-xxl/config.json	
read	tests/assets/sft_test/data.json	
read	tests/assets/synthetic_multimodal/generate.py	
read	tests/assets/tokenizer/tokenizer.json	
read	tests/assets/tokenizer/tokenizer_config.json	
read	tests/integration_tests/__init__.py	
read	tests/integration_tests/b200.py	
read	tests/integration_tests/features.py	
read	tests/integration_tests/flux.py	
read	tests/integration_tests/h100.py	
read	tests/integration_tests/models.py	
read	tests/integration_tests/run_tests.py	
read	tests/rl/__init__.py	
read	tests/rl/integration_tests/__init__.py	
read	tests/rl/integration_tests/rl.py	
read	tests/rl/unit_tests/__init__.py	
read	tests/rl/unit_tests/cpu/__init__.py	
read	tests/rl/unit_tests/cpu/test_alphabet_sort.py	
read	tests/rl/unit_tests/cpu/test_async_controller.py	
read	tests/rl/unit_tests/cpu/test_batch_invariance.py	
read	tests/rl/unit_tests/cpu/test_batcher.py	
read	tests/rl/unit_tests/cpu/test_bin_scheduler.py	
read	tests/rl/unit_tests/cpu/test_dapo_math.py	
read	tests/rl/unit_tests/cpu/test_engine_loop.py	
read	tests/rl/unit_tests/cpu/test_generator.py	
read	tests/rl/unit_tests/cpu/test_inter_generator_router.py	
read	tests/rl/unit_tests/cpu/test_intra_generator_router.py	
read	tests/rl/unit_tests/cpu/test_losses.py	
read	tests/rl/unit_tests/cpu/test_metrics.py	
read	tests/rl/unit_tests/cpu/test_muse_glimmer_renderer.py	

Event	Path or command	Hash
read	tests/rl/unit_tests/cpu/test_policy_loss_configs.py	
read	tests/rl/unit_tests/cpu/test_rollout_recorder.py	
read	tests/rl/unit_tests/cpu/test_rollout_utils.py	
read	tests/rl/unit_tests/cpu/test_rollout_worker.py	
read	tests/rl/unit_tests/cpu/test_rollout_pool.py	
read	tests/rl/unit_tests/cpu/test_runtime_env.py	
read	tests/rl/unit_tests/cpu/test_search_r1.py	
read	tests/rl/unit_tests/cpu/test_shutdown.py	
read	tests/rl/unit_tests/cpu/test_token_env.py	
read	tests/rl/unit_tests/cpu/test_train_loop.py	
read	tests/rl/unit_tests/cpu/test_trainer.py	
read	tests/rl/unit_tests/cpu/test_verifiers.py	
read	tests/rl/unit_tests/cpu/test_verifiers_example.py	
read	tests/rl/unit_tests/cpu/test_vllm_otel_stat_logger.py	
read	tests/rl/unit_tests/cpu/test_vllm_registry.py	
read	tests/rl/unit_tests/cpu/test_vllm_wrapper.py	
read	tests/rl/unit_tests/cpu/test_weight_sync.py	
read	tests/rl/unit_tests/gpu/__init__.py	
read	tests/rl/unit_tests/gpu/test_bitwise_parity.py	
read	tests/rl/unit_tests/gpu/test_gdn_cudagraph.py	
read	tests/rl/unit_tests/gpu/test_gdn_graph_metadata.py	
read	tests/rl/unit_tests/gpu/test_generator_uneven_decode.py	
read	tests/unit_tests/__init__.py	
read	tests/unit_tests/cpu/__init__.py	
read	tests/unit_tests/cpu/components/data/test_grain_data.py	
read	tests/unit_tests/cpu/components/data/test_qwen_multimodal_data.py	
read	tests/unit_tests/cpu/components/data/test_synthetic_multimodal_data.py	
read	tests/unit_tests/cpu/flex_shard/__init__.py	
read	tests/unit_tests/cpu/flex_shard/test_dist_muon_storage_validation.py	
read	tests/unit_tests/cpu/flex_shard/test_optimizer_reshard_runtime.py	
read	tests/unit_tests/cpu/flex_shard/test_optimizer_reshard_schedule.py	
read	tests/unit_tests/cpu/observability/__init__.py	
read	tests/unit_tests/cpu/observability/test_logging.py	
read	tests/unit_tests/cpu/observability/test_metrics_config.py	
read	tests/unit_tests/cpu/observability/test_metrics_rank.py	
read	tests/unit_tests/cpu/observability/test_profiler.py	
read	tests/unit_tests/cpu/observability/test_profiler_config.py	
read	tests/unit_tests/cpu/observability/test_sdc_replayer.py	
read	tests/unit_tests/cpu/observability/test_structured_logging.py	
read	tests/unit_tests/cpu/observability/test_validation_metrics_window.py	
read	tests/unit_tests/cpu/observability/test_wandb_logger.py	
read	tests/unit_tests/cpu/test_aux_loss.py	
read	tests/unit_tests/cpu/test_batch_invariance.py	
read	tests/unit_tests/cpu/test_chat_dataset.py	
read	tests/unit_tests/cpu/test_checkpoint.py	
read	tests/unit_tests/cpu/test_compile_config.py	
read	tests/unit_tests/cpu/test_config_converter.py	
read	tests/unit_tests/cpu/test_config_loader.py	
read	tests/unit_tests/cpu/test_config_validation.py	
read	tests/unit_tests/cpu/test_configurable.py	

Event	Path or command	Hash
read	tests/unit_tests/cpu/test_context_parallel_validation.py	
read	tests/unit_tests/cpu/test_convert_ema_to_hf.py	
read	tests/unit_tests/cpu/test_cp_attention.py	
read	tests/unit_tests/cpu/test_cuda_graph.py	
read	tests/unit_tests/cpu/test_dataset_checkpointing.py	
read	tests/unit_tests/cpu/test_dataset_flux.py	
read	tests/unit_tests/cpu/test_debug_config_defaults.py	
read	tests/unit_tests/cpu/test_deepseek_v3_moe.py	
read	tests/unit_tests/cpu/test_deepseek_v4_compressor.py	
read	tests/unit_tests/cpu/test_deepseek_v4_dsa.py	
read	tests/unit_tests/cpu/test_deepseek_v4_flops.py	
read	tests/unit_tests/cpu/test_deepseek_v4_moe.py	
read	tests/unit_tests/cpu/test_deepseek_v4_mtp.py	
read	tests/unit_tests/cpu/test_dist_moe.py	
read	tests/unit_tests/cpu/test_distributed_utils.py	
read	tests/unit_tests/cpu/test_download_hf_assets.py	
read	tests/unit_tests/cpu/test_ema.py	
read	tests/unit_tests/cpu/test_ema_resharding.py	
read	tests/unit_tests/cpu/test_embedding.py	
read	tests/unit_tests/cpu/test_expert_parallel.py	
read	tests/unit_tests/cpu/test_feed_forward.py	
read	tests/unit_tests/cpu/test_filesystem.py	
read	tests/unit_tests/cpu/test_flex_attention.py	
read	tests/unit_tests/cpu/test_flux_inference.py	
read	tests/unit_tests/cpu/test_flux_mxfp8_readme.py	
read	tests/unit_tests/cpu/test_flux_sharding.py	
read	tests/unit_tests/cpu/test_fsdp.py	
read	tests/unit_tests/cpu/test_fused_mla_override.py	
read	tests/unit_tests/cpu/test_gated_rmsnorm.py	
read	tests/unit_tests/cpu/test_gc_freq.py	
read	tests/unit_tests/cpu/test_gpt_oss_moe.py	
read	tests/unit_tests/cpu/test_hi_mid_lo_linear.py	
read	tests/unit_tests/cpu/test_inference_moe.py	
read	tests/unit_tests/cpu/test_integration_test_definitions.py	
read	tests/unit_tests/cpu/test_invalid_loss.py	
read	tests/unit_tests/cpu/test_kimi_k3_pp_block_grads.py	
read	tests/unit_tests/cpu/test_kimi_k3_pp_layout.py	
read	tests/unit_tests/cpu/test_kimi_k3_pp_stage.py	
read	tests/unit_tests/cpu/test_linear.py	
read	tests/unit_tests/cpu/test_lora.py	
read	tests/unit_tests/cpu/test_loss.py	
read	tests/unit_tests/cpu/test_lr_scheduler.py	
read	tests/unit_tests/cpu/test_mm_dataset_checkpointing.py	
read	tests/unit_tests/cpu/test_mm_dataset_preprocessing.py	
read	tests/unit_tests/cpu/test_model_td_layout.py	
read	tests/unit_tests/cpu/test_module.py	
read	tests/unit_tests/cpu/test_moe.py	
read	tests/unit_tests/cpu/test_multimodal_audio_utils.py	
read	tests/unit_tests/cpu/test_muse_glimmer.py	
read	tests/unit_tests/cpu/test_offloading.py	

Event	Path or command	Hash
read	tests/unit_tests/cpu/test_offset_rmsnorm.py	
read	tests/unit_tests/cpu/test_optim.py	
read	tests/unit_tests/cpu/test_optimizer_param_groups.py	
read	tests/unit_tests/cpu/test_optional_imports.py	
read	tests/unit_tests/cpu/test_override.py	
read	tests/unit_tests/cpu/test_packed_vision.py	
read	tests/unit_tests/cpu/test_parallelism_config.py	
read	tests/unit_tests/cpu/test_parallelism_context.py	
read	tests/unit_tests/cpu/test_pipeline_parallel.py	
read	tests/unit_tests/cpu/test_qkv_linear.py	
read	tests/unit_tests/cpu/test_quantization.py	
read	tests/unit_tests/cpu/test_quantized_fsdp_tensor.py	
read	tests/unit_tests/cpu/test_qwen3_5.py	
read	tests/unit_tests/cpu/test_qwen3_5_moe.py	
read	tests/unit_tests/cpu/test_qwen3_5_mrope_positions.py	
read	tests/unit_tests/cpu/test_qwen3_6.py	
read	tests/unit_tests/cpu/test_qwen3_8.py	
read	tests/unit_tests/cpu/test_renderer.py	
read	tests/unit_tests/cpu/test_rmsnorm.py	
read	tests/unit_tests/cpu/test_rope.py	
read	tests/unit_tests/cpu/test_set_determinism.py	
read	tests/unit_tests/cpu/test_skip_dp.py	
read	tests/unit_tests/cpu/test_state_dict_adapter.py	
read	tests/unit_tests/cpu/test_state_dict_keys.py	
read	tests/unit_tests/cpu/test_synthetic_multimodal_fixtures.py	
read	tests/unit_tests/cpu/test_text_dataset_packing.py	
read	tests/unit_tests/cpu/test_tokenizer.py	
read	tests/unit_tests/cpu/test_tools_utils.py	
read	tests/unit_tests/cpu/test_torch_checkpointing.py	
read	tests/unit_tests/cpu/test_torchft_trainer.py	
read	tests/unit_tests/cpu/test_tp_kv_heads_validation.py	
read	tests/unit_tests/cpu/test_train_spec.py	
read	tests/unit_tests/cpu/test_trainer.py	
read	tests/unit_tests/cpu/test_transforms.py	
read	tests/unit_tests/cpu/test_validate.py	
read	tests/unit_tests/cpu/test_validator_frequency.py	
read	tests/unit_tests/cpu/test_varlen_attention.py	
read	tests/unit_tests/cpu/test_vision_attention.py	
read	tests/unit_tests/cpu/test_weight_tying.py	
read	tests/unit_tests/gpu/__init__.py	
read	tests/unit_tests/gpu/flex_shard/__init__.py	
read	tests/unit_tests/gpu/flex_shard/test_dist_muon.py	
read	tests/unit_tests/gpu/test_activation_checkpoint.py	
read	tests/unit_tests/gpu/test_async_linear.py	
read	tests/unit_tests/gpu/test_cuda_graph_group_replay.py	
read	tests/unit_tests/gpu/test_deepest_activation_checkpoint.py	
read	tests/unit_tests/gpu/test_deepseek_v4_attention.py	
read	tests/unit_tests/gpu/test_dist_moe.py	
read	tests/unit_tests/gpu/test_distributed_linear.py	
read	tests/unit_tests/gpu/test_ema.py	

Event	Path or command	Hash
read	tests/unit_tests/gpu/test_fsdp_cpu_offload.py	
read	tests/unit_tests/gpu/test_fsdp_embedding.py	
read	tests/unit_tests/gpu/test_fsdp_moe_sharding.py	
read	tests/unit_tests/gpu/test_fused_swiglu.py	
read	tests/unit_tests/gpu/test_fused_swiglu_override.py	
read	tests/unit_tests/gpu/test_gated_rmsnorm.py	
read	tests/unit_tests/gpu/test_helion_rope.py	
read	tests/unit_tests/gpu/test_hi_mid_lo_linear.py	
read	tests/unit_tests/gpu/test_kda_attention.py	
read	tests/unit_tests/gpu/test_kimi_k3.py	
read	tests/unit_tests/gpu/test_mxfp8_fsdp.py	
read	tests/unit_tests/gpu/test_mxfp8_linear.py	
read	tests/unit_tests/gpu/test_nvfp4_fsdp.py	
read	tests/unit_tests/gpu/test_nvfp4_linear.py	
read	tests/unit_tests/gpu/test_offset_rmsnorm.py	
read	tests/unit_tests/gpu/test_qk_clip.py	
read	tests/unit_tests/gpu/test_quantile_balance.py	
read	tests/unit_tests/gpu/test_qwen3_5_deltanet.py	
read	tests/unit_tests/gpu/test_remat.py	
read	tests/unit_tests/gpu/test_remat_all_to_all.py	
read	tests/unit_tests/gpu/test_rope_compile.py	
read	tests/unit_tests/gpu/test_sigmoid_gate.py	
read	tests/unit_tests/gpu/test_situglu.py	
read	tests/unit_tests/gpu/test_swiglu.py	
read	tests/unit_tests/gpu/test_tensor_parallel_feed_forward.py	
read	tests/utils.py	
read	torchtitan/__init__.py	
read	torchtitan/components/checkpointer/__init__.py	
read	torchtitan/components/checkpointer/base.py	
read	torchtitan/components/checkpointer/dcp.py	
read	torchtitan/components/checkpointer/torch_checkpointing.py	
read	torchtitan/components/checkpointer/utils.py	
read	torchtitan/components/data/__init__.py	
read	torchtitan/components/data/collators.py	
read	torchtitan/components/data/dataset.py	
read	torchtitan/components/data/loader.py	
read	torchtitan/components/data/packing.py	
read	torchtitan/components/data/sources.py	
read	torchtitan/components/data/types.py	
read	torchtitan/components/loss.py	
read	torchtitan/components/optim/__init__.py	
read	torchtitan/components/optim/ema.py	
read	torchtitan/components/optim/lr_scheduler.py	
read	torchtitan/components/optim/optim.py	
read	torchtitan/components/optim/optimizer.py	
read	torchtitan/components/optim/utils.py	
read	torchtitan/components/renderer.py	
read	torchtitan/components/tokenizer.py	
read	torchtitan/components/validate.py	
read	torchtitan/config/__init__.py	

Event	Path or command	Hash
read	torchtitan/config/configs.py	
read	torchtitan/config/configurable.py	
read	torchtitan/config/function.py	
read	torchtitan/config/loader.py	
read	torchtitan/config/override.py	
read	torchtitan/config/parallelism.py	
read	torchtitan/config/transform/__init__.py	
read	torchtitan/config/transform/apply.py	
read	torchtitan/config/transform/async_tensor_parallel.py	
read	torchtitan/config/transform/base.py	
read	torchtitan/config/transform/batch_invariance.py	
read	torchtitan/config/transform/context_parallel.py	
read	torchtitan/config/transform/converter.py	
read	torchtitan/config/transform/dist_moe.py	
read	torchtitan/config/transform/lm_head_fp32.py	
read	torchtitan/config/transform/lora.py	
read	torchtitan/config/transform/quantization.py	
read	torchtitan/config/transform/relations.py	
read	torchtitan/config/transform/token_dispatcher.py	
read	torchtitan/config/validation.py	
read	torchtitan/distributed/__init__.py	
read	torchtitan/distributed/activation_checkpoint.py	
read	torchtitan/distributed/batch_invariant.py	
read	torchtitan/distributed/context_parallel.py	
read	torchtitan/distributed/cuda_graph.py	
read	torchtitan/distributed/deepep/deepep.py	
read	torchtitan/distributed/deepep/hybridep.py	
read	torchtitan/distributed/flex_shard/__init__.py	
read	torchtitan/distributed/flex_shard/_optimizer_reshard_runtime.py	
read	torchtitan/distributed/flex_shard/_optimizer_reshard_schedule.py	
read	torchtitan/distributed/flex_shard/dist_muon.py	
read	torchtitan/distributed/flex_shard/optimizer_reshard.py	
read	torchtitan/distributed/fsdp.py	
read	torchtitan/distributed/local_compile.py	
read	torchtitan/distributed/offloading.py	
read	torchtitan/distributed/parallelism_context.py	
read	torchtitan/distributed/pipeline_parallel.py	
read	torchtitan/distributed/spmd_types.py	
read	torchtitan/distributed/utils.py	
read	torchtitan/experiments/__init__.py	
read	torchtitan/experiments/graph_trainer/__init__.py	
read	torchtitan/experiments/graph_trainer/autoparallel_api.py	
read	torchtitan/experiments/graph_trainer/autoresearch/scripts/run_benchmark.sh	
read	torchtitan/experiments/graph_trainer/chunked_loss.py	
read	torchtitan/experiments/graph_trainer/common_utils.py	
read	torchtitan/experiments/graph_trainer/compile.py	
read	torchtitan/experiments/graph_trainer/configs.py	
read	torchtitan/experiments/graph_trainer/cpu_offload.py	
read	torchtitan/experiments/graph_trainer/cuda_graph.py	
read	torchtitan/experiments/graph_trainer/custom_codegen.py	

Event	Path or command	Hash
read	torchtitan/experiments/graph_trainer/debug_utils.py	
read	torchtitan/experiments/graph_trainer/decompositions.py	
read	torchtitan/experiments/graph_trainer/deepseek_v3/__init__.py	
read	torchtitan/experiments/graph_trainer/deepseek_v3/flavors.py	
read	torchtitan/experiments/graph_trainer/deepseek_v3/model.py	
read	torchtitan/experiments/graph_trainer/deepseek_v3/parallelize_autoparallel.py	
read	torchtitan/experiments/graph_trainer/dynamic_shapes.py	
read	torchtitan/experiments/graph_trainer/ep_eager_chunk.py	
read	torchtitan/experiments/graph_trainer/ep_overlap_pass.py	
read	torchtitan/experiments/graph_trainer/ep_pass_utils.py	
read	torchtitan/experiments/graph_trainer/ep_process_group_pass.py	
read	torchtitan/experiments/graph_trainer/fsdp_passes.py	
read	torchtitan/experiments/graph_trainer/fsdp_patterns.py	
read	torchtitan/experiments/graph_trainer/grad_accumulation.py	
read	torchtitan/experiments/graph_trainer/graph_builder.py	
read	torchtitan/experiments/graph_trainer/graph_builder_utils.py	
read	torchtitan/experiments/graph_trainer/graph_pp/__init__.py	
read	torchtitan/experiments/graph_trainer/graph_pp/graph_mux.py	
read	torchtitan/experiments/graph_trainer/graph_pp/partition.py	
read	torchtitan/experiments/graph_trainer/graph_pp/pipeline.py	
read	torchtitan/experiments/graph_trainer/graph_pp/pp_graph_builder.py	
read	torchtitan/experiments/graph_trainer/graph_pp/runner.py	
read	torchtitan/experiments/graph_trainer/graph_pp/split_ddp.py	
read	torchtitan/experiments/graph_trainer/graph_pp/split_fsdp_collectives.py	
read	torchtitan/experiments/graph_trainer/graph_pp/stage.py	
read	torchtitan/experiments/graph_trainer/graph_pp/stage_builder.py	
read	torchtitan/experiments/graph_trainer/graph_pp/utils.py	
read	torchtitan/experiments/graph_trainer/inductor_passes.py	
read	torchtitan/experiments/graph_trainer/llama3/__init__.py	
read	torchtitan/experiments/graph_trainer/llama3/flavors.py	
read	torchtitan/experiments/graph_trainer/llama3/model.py	
read	torchtitan/experiments/graph_trainer/llama3/parallelize_autoparallel.py	
read	torchtitan/experiments/graph_trainer/log_activation_memory_policy.py	
read	torchtitan/experiments/graph_trainer/make_fx_tracer.py	
read	torchtitan/experiments/graph_trainer/memory_policy.py	
read	torchtitan/experiments/graph_trainer/model.py	
read	torchtitan/experiments/graph_trainer/muse_glimmer/__init__.py	
read	torchtitan/experiments/graph_trainer/muse_glimmer/flavors.py	
read	torchtitan/experiments/graph_trainer/muse_glimmer/model.py	
read	torchtitan/experiments/graph_trainer/mutation_utils.py	
read	torchtitan/experiments/graph_trainer/paged_stash_memory_policy.py	
read	torchtitan/experiments/graph_trainer/passes.py	
read	torchtitan/experiments/graph_trainer/performance_passes.py	
read	torchtitan/experiments/graph_trainer/precompile.py	
read	torchtitan/experiments/graph_trainer/precompile_main.py	
read	torchtitan/experiments/graph_trainer/qwen3/__init__.py	
read	torchtitan/experiments/graph_trainer/qwen3/flavors.py	
read	torchtitan/experiments/graph_trainer/qwen3/model.py	
read	torchtitan/experiments/graph_trainer/registry.py	
read	torchtitan/experiments/graph_trainer/remove_noop_passes.py	

Event	Path or command	Hash
read	torchtitan/experiments/graph_trainer/run_train_precompile.sh	
read	torchtitan/experiments/graph_trainer/selective_activation_remat.py	
read	torchtitan/experiments/graph_trainer/simple_fsdp.py	
read	torchtitan/experiments/graph_trainer/spmd_gradient_accumulation_graph_bui...	
read	torchtitan/experiments/graph_trainer/spmd_graph_builder.py	
read	torchtitan/experiments/graph_trainer/storage.py	
read	torchtitan/experiments/graph_trainer/subgraph_regions.py	
read	torchtitan/experiments/graph_trainer/tests/__init__.py	
read	torchtitan/experiments/graph_trainer/tests/_trainer_test_utils.py	
read	torchtitan/experiments/graph_trainer/tests/integration_tests.py	
read	torchtitan/experiments/graph_trainer/tests/run_precompile_tests.py	
read	torchtitan/experiments/graph_trainer/tests/test_autoparallel.py	
read	torchtitan/experiments/graph_trainer/tests/test_bitwise_deterministic.py	
read	torchtitan/experiments/graph_trainer/tests/test_chunked_loss.py	
read	torchtitan/experiments/graph_trainer/tests/test_cpu_offload.py	
read	torchtitan/experiments/graph_trainer/tests/test_custom_codegen.py	
read	torchtitan/experiments/graph_trainer/tests/test_dist_moe.py	
read	torchtitan/experiments/graph_trainer/tests/test_grad_accumulation.py	
read	torchtitan/experiments/graph_trainer/tests/test_graph_pp_passes.py	
read	torchtitan/experiments/graph_trainer/tests/test_graph_pp_runner.py	
read	torchtitan/experiments/graph_trainer/tests/test_numerics.py	
read	torchtitan/experiments/graph_trainer/tests/test_paged_stash.py	
read	torchtitan/experiments/graph_trainer/tests/test_passes.py	
read	torchtitan/experiments/graph_trainer/tests/test_performance_passes.py	
read	torchtitan/experiments/graph_trainer/tests/test_precompile.py	
read	torchtitan/experiments/graph_trainer/tests/test_profiler.py	
read	torchtitan/experiments/graph_trainer/tests/test_remat_metadata.py	
read	torchtitan/experiments/graph_trainer/tests/test_sac_peak_memory.py	
read	torchtitan/experiments/graph_trainer/tests/test_simple_fsdp.py	
read	torchtitan/experiments/graph_trainer/tests/test_subgraph_regions.py	
read	torchtitan/experiments/graph_trainer/tests/test_trace_module.py	
read	torchtitan/experiments/graph_trainer/tests/test_wgrad_accumulation.py	
read	torchtitan/experiments/graph_trainer/trainer.py	
read	torchtitan/experiments/graph_trainer/wgrad_accumulation.py	
read	torchtitan/experiments/torchft/__init__.py	
read	torchtitan/experiments/torchft/checkpoint.py	
read	torchtitan/experiments/torchft/config/__init__.py	
read	torchtitan/experiments/torchft/config/job_config.py	
read	torchtitan/experiments/torchft/diloco/__init__.py	
read	torchtitan/experiments/torchft/diloco/utils.py	
read	torchtitan/experiments/torchft/llama3/__init__.py	
read	torchtitan/experiments/torchft/llama3/flavors.py	
read	torchtitan/experiments/torchft/llama3/model.py	
read	torchtitan/experiments/torchft/manager.py	
read	torchtitan/experiments/torchft/optimizer.py	
read	torchtitan/experiments/torchft/tests/__init__.py	
read	torchtitan/experiments/torchft/tests/integration_tests.py	
read	torchtitan/experiments/torchft/tests/test_torchft_checkpoint.py	
read	torchtitan/experiments/torchft/tests/test_torchft_manager.py	
read	torchtitan/experiments/torchft/tests/test_torchft_optimizer.py	

Event	Path or command	Hash
read	torchtitan/experiments/torchft/trainer.py	
read	torchtitan/experiments/transformers_modeling_backend/.claude/skills/add_m...	
read	torchtitan/experiments/transformers_modeling_backend/.claude/skills/add_m...	
read	torchtitan/experiments/transformers_modeling_backend/__init__.py	
read	torchtitan/experiments/transformers_modeling_backend/flavors.py	
read	torchtitan/experiments/transformers_modeling_backend/hf_sharding.py	
read	torchtitan/experiments/transformers_modeling_backend/model.py	
read	torchtitan/experiments/transformers_modeling_backend/module_conversion.py	
read	torchtitan/experiments/transformers_modeling_backend/moe_replacement.py	
read	torchtitan/experiments/transformers_modeling_backend/parallelize.py	
read	torchtitan/experiments/transformers_modeling_backend/pipeline.py	
read	torchtitan/experiments/transformers_modeling_backend/state_dict_adapter.py	
read	torchtitan/experiments/transformers_modeling_backend/tests/__init__.py	
read	torchtitan/experiments/transformers_modeling_backend/tests/cp_pp_numeric...	
read	torchtitan/experiments/transformers_modeling_backend/tests/integration_test...	
read	torchtitan/experiments/transformers_modeling_backend/tests/run_tests.sh	
read	torchtitan/experiments/transformers_modeling_backend/tests/test_flex_attenti...	
read	torchtitan/experiments/transformers_modeling_backend/tests/test_flex_cp_nu...	
read	torchtitan/experiments/transformers_modeling_backend/tests/test_hf_build_f...	
read	torchtitan/experiments/transformers_modeling_backend/tests/test_hf_shardin...	
read	torchtitan/experiments/transformers_modeling_backend/tests/test_moe_parall...	
read	torchtitan/experiments/transformers_modeling_backend/tokenizer.py	
read	torchtitan/hf_datasets/__init__.py	
read	torchtitan/hf_datasets/multimodal/__init__.py	
read	torchtitan/hf_datasets/multimodal/mm_collator.py	
read	torchtitan/hf_datasets/multimodal/mm_datasets.py	
read	torchtitan/hf_datasets/multimodal/utils/audio.py	
read	torchtitan/hf_datasets/multimodal/utils/image.py	
read	torchtitan/hf_datasets/multimodal/utils/text.py	
read	torchtitan/hf_datasets/multimodal/utils/video.py	
read	torchtitan/hf_datasets/text_datasets.py	
read	torchtitan/models/__init__.py	
read	torchtitan/models/common/__init__.py	
read	torchtitan/models/common/activation.py	
read	torchtitan/models/common/async_linear.py	
read	torchtitan/models/common/attention/__init__.py	
read	torchtitan/models/common/attention/attention.py	
read	torchtitan/models/common/attention/cp_attention.py	
read	torchtitan/models/common/attention/cp_kda.py	
read	torchtitan/models/common/attention/kda.py	
read	torchtitan/models/common/aux_loss.py	
read	torchtitan/models/common/config_utils.py	
read	torchtitan/models/common/decoder.py	
read	torchtitan/models/common/decoder_sharding.py	
read	torchtitan/models/common/dist_moe/__init__.py	
read	torchtitan/models/common/dist_moe/mx8.py	
read	torchtitan/models/common/dist_moe/routed_experts.py	
read	torchtitan/models/common/dist_moe/runtime.py	
read	torchtitan/models/common/embedding.py	
read	torchtitan/models/common/feed_forward.py	

Event	Path or command	Hash
read	torchtitan/models/common/hi_mid_lo_linear.py	
read	torchtitan/models/common/linear.py	
read	torchtitan/models/common/lora.py	
read	torchtitan/models/common/moe.py	
read	torchtitan/models/common/moe_sharding.py	
read	torchtitan/models/common/multimodal.py	
read	torchtitan/models/common/nn_modules.py	
read	torchtitan/models/common/norm.py	
read	torchtitan/models/common/param_init.py	
read	torchtitan/models/common/rope.py	
read	torchtitan/models/common/token_dispatcher.py	
read	torchtitan/models/common/vision_encoder.py	
read	torchtitan/models/common/vision_encoder_sharding.py	
read	torchtitan/models/deepseek_v3/__init__.py	
read	torchtitan/models/deepseek_v3/flavors.py	
read	torchtitan/models/deepseek_v3/model.py	
read	torchtitan/models/deepseek_v3/moe.py	
read	torchtitan/models/deepseek_v3/mtp.py	
read	torchtitan/models/deepseek_v3/sharding.py	
read	torchtitan/models/deepseek_v3/state_dict_adapter.py	
read	torchtitan/models/deepseek_v4/__init__.py	
read	torchtitan/models/deepseek_v4/attention.py	
read	torchtitan/models/deepseek_v4/compressor.py	
read	torchtitan/models/deepseek_v4/flavors.py	
read	torchtitan/models/deepseek_v4/mhc.py	
read	torchtitan/models/deepseek_v4/model.py	
read	torchtitan/models/deepseek_v4/moe.py	
read	torchtitan/models/deepseek_v4/mtp.py	
read	torchtitan/models/deepseek_v4/sharding.py	
read	torchtitan/models/deepseek_v4/state_dict_adapter.py	
read	torchtitan/models/flux/__init__.py	
read	torchtitan/models/flux/configs.py	
read	torchtitan/models/flux/flavors.py	
read	torchtitan/models/flux/flux_datasets.py	
read	torchtitan/models/flux/inference/infer.py	
read	torchtitan/models/flux/inference/sampling.py	
read	torchtitan/models/flux/model/autoencoder.py	
read	torchtitan/models/flux/model/hf_embedder.py	
read	torchtitan/models/flux/model/layers.py	
read	torchtitan/models/flux/model/model.py	
read	torchtitan/models/flux/model/state_dict_adapter.py	
read	torchtitan/models/flux/run_infer.sh	
read	torchtitan/models/flux/sharding.py	
read	torchtitan/models/flux/tokenizer.py	
read	torchtitan/models/flux/trainer.py	
read	torchtitan/models/flux/utils.py	
read	torchtitan/models/flux/validate.py	
read	torchtitan/models/gpt_oss/__init__.py	
read	torchtitan/models/gpt_oss/flavors.py	
read	torchtitan/models/gpt_oss/model.py	

Event	Path or command	Hash
read	torchtitan/models/gpt_oss/moe.py	
read	torchtitan/models/gpt_oss/sharding.py	
read	torchtitan/models/gpt_oss/state_dict_adapter.py	
read	torchtitan/models/kimi_k2_7/__init__.py	
read	torchtitan/models/kimi_k2_7/flavors.py	
read	torchtitan/models/kimi_k2_7/model.py	
read	torchtitan/models/kimi_k2_7/qk_clip.py	
read	torchtitan/models/kimi_k2_7/sharding.py	
read	torchtitan/models/kimi_k2_7/state_dict_adapter.py	
read	torchtitan/models/kimi_k2_7/vision_encoder.py	
read	torchtitan/models/kimi_k3/__init__.py	
read	torchtitan/models/kimi_k3/flavors.py	
read	torchtitan/models/kimi_k3/model.py	
read	torchtitan/models/kimi_k3/moe.py	
read	torchtitan/models/kimi_k3/pipeline_parallel/__init__.py	
read	torchtitan/models/kimi_k3/pipeline_parallel/cache.py	
read	torchtitan/models/kimi_k3/pipeline_parallel/layout.py	
read	torchtitan/models/kimi_k3/pipeline_parallel/stage.py	
read	torchtitan/models/kimi_k3/sharding.py	
read	torchtitan/models/kimi_k3/state_dict_adapter.py	
read	torchtitan/models/kimi_k3/vision_encoder.py	
read	torchtitan/models/llama3/__init__.py	
read	torchtitan/models/llama3/flavors.py	
read	torchtitan/models/llama3/model.py	
read	torchtitan/models/llama3/sharding.py	
read	torchtitan/models/llama3/state_dict_adapter.py	
read	torchtitan/models/muse_glimmer/__init__.py	
read	torchtitan/models/muse_glimmer/flavors.py	
read	torchtitan/models/muse_glimmer/model.py	
read	torchtitan/models/muse_glimmer/sharding.py	
read	torchtitan/models/muse_glimmer/state_dict_adapter.py	
read	torchtitan/models/muse_glimmer/vision_encoder.py	
read	torchtitan/models/qwen3/__init__.py	
read	torchtitan/models/qwen3/flavors.py	
read	torchtitan/models/qwen3/model.py	
read	torchtitan/models/qwen3/sharding.py	
read	torchtitan/models/qwen3/state_dict_adapter.py	
read	torchtitan/models/qwen3_5/__init__.py	
read	torchtitan/models/qwen3_5/flavors.py	
read	torchtitan/models/qwen3_5/gdn.py	
read	torchtitan/models/qwen3_5/model.py	
read	torchtitan/models/qwen3_5/moe.py	
read	torchtitan/models/qwen3_5/rope.py	
read	torchtitan/models/qwen3_5/sharding.py	
read	torchtitan/models/qwen3_5/state_dict_adapter.py	
read	torchtitan/models/qwen3_5/vision_encoder.py	
read	torchtitan/models/qwen3_6/__init__.py	
read	torchtitan/models/qwen3_6/flavors.py	
read	torchtitan/models/qwen3_8/__init__.py	
read	torchtitan/models/qwen3_8/flavors.py	

Event	Path or command	Hash
read	torchtitan/models/utils.py	
read	torchtitan/observability/__init__.py	
read	torchtitan/observability/logging.py	
read	torchtitan/observability/metrics.py	
read	torchtitan/observability/profiler.py	
read	torchtitan/observability/sdc_replayer.py	
read	torchtitan/observability/structured_logger/__init__.py	
read	torchtitan/observability/structured_logger/gantt_generator.py	
read	torchtitan/observability/structured_logger/jsonl_handler.py	
read	torchtitan/observability/structured_logger/step_state.py	
read	torchtitan/observability/structured_logger/structured_logging.py	
read	torchtitan/ops/__init__.py	
read	torchtitan/ops/scatter_add.py	
read	torchtitan/protocols/__init__.py	
read	torchtitan/protocols/model.py	
read	torchtitan/protocols/module.py	
read	torchtitan/protocols/sharding.py	
read	torchtitan/protocols/state_dict_adapter.py	
read	torchtitan/quantization/__init__.py	
read	torchtitan/quantization/_fsdp_tensor.py	
read	torchtitan/quantization/mxfp8/__init__.py	
read	torchtitan/quantization/mxfp8/experts.py	
read	torchtitan/quantization/mxfp8/linear.py	
read	torchtitan/quantization/mxfp8/tensor.py	
read	torchtitan/quantization/nvfp4/__init__.py	
read	torchtitan/quantization/nvfp4/experts.py	
read	torchtitan/quantization/nvfp4/linear.py	
read	torchtitan/quantization/nvfp4/tensor.py	
read	torchtitan/quantization/nvfp4/utils.py	
read	torchtitan/quantization/utils.py	
read	torchtitan/rl/__init__.py	
read	torchtitan/rl/_runtime.py	
read	torchtitan/rl/components/__init__.py	
read	torchtitan/rl/components/batcher.py	
read	torchtitan/rl/components/training_sample_builder.py	
read	torchtitan/rl/components/work_buffer.py	
read	torchtitan/rl/controller.py	
read	torchtitan/rl/distributed/__init__.py	
read	torchtitan/rl/distributed/actors/__init__.py	
read	torchtitan/rl/distributed/actors/generator.py	
read	torchtitan/rl/distributed/actors/rollout_worker.py	
read	torchtitan/rl/distributed/actors/trainer.py	
read	torchtitan/rl/distributed/parallelism.py	
read	torchtitan/rl/distributed/routing/__init__.py	
read	torchtitan/rl/distributed/routing/inter_generator.py	
read	torchtitan/rl/distributed/routing/intra_generator.py	
read	torchtitan/rl/distributed/routing/strategies.py	
read	torchtitan/rl/distributed/routing/types.py	
read	torchtitan/rl/distributed/torch_elastic.py	
read	torchtitan/rl/distributed/weight_sync.py	

Event	Path or command	Hash
read	torchtitan/rl/examples/__init__.py	
read	torchtitan/rl/examples/alphabet_sort/__init__.py	
read	torchtitan/rl/examples/alphabet_sort/data.py	
read	torchtitan/rl/examples/alphabet_sort/env.py	
read	torchtitan/rl/examples/alphabet_sort/rubric.py	
read	torchtitan/rl/examples/dapo_math/__init__.py	
read	torchtitan/rl/examples/dapo_math/data.py	
read	torchtitan/rl/examples/dapo_math/env.py	
read	torchtitan/rl/examples/dapo_math/rubric.py	
read	torchtitan/rl/examples/dapo_math/thread_timeout.py	
read	torchtitan/rl/examples/search_r1/__init__.py	
read	torchtitan/rl/examples/search_r1/data.py	
read	torchtitan/rl/examples/search_r1/env.py	
read	torchtitan/rl/examples/search_r1/rubric.py	
read	torchtitan/rl/examples/verifiers/__init__.py	
read	torchtitan/rl/examples/verifiers/dapo_math/__init__.py	
read	torchtitan/rl/examples/verifiers/dapo_math/data.py	
read	torchtitan/rl/examples/verifiers/data.py	
read	torchtitan/rl/examples/verifiers/env_server.py	
read	torchtitan/rl/examples/verifiers/generation_server.py	
read	torchtitan/rl/examples/verifiers/rollouter.py	
read	torchtitan/rl/generate.py	
read	torchtitan/rl/generator.py	
read	torchtitan/rl/losses/__init__.py	
read	torchtitan/rl/losses/dapo.py	
read	torchtitan/rl/losses/grpo.py	
read	torchtitan/rl/model/attention.py	
read	torchtitan/rl/model/batch_invariance.py	
read	torchtitan/rl/model/gdn.py	
read	torchtitan/rl/model/kda.py	
read	torchtitan/rl/model/linear_attention_backend.py	
read	torchtitan/rl/model/muse_glimmer/__init__.py	
read	torchtitan/rl/model/muse_glimmer/atem.py	
read	torchtitan/rl/model/muse_glimmer/renderer.py	
read	torchtitan/rl/model/vllm_registry.py	
read	torchtitan/rl/model/vllm_worker.py	
read	torchtitan/rl/model/vllm_wrapper.py	
read	torchtitan/rl/observability/__init__.py	
read	torchtitan/rl/observability/controller.py	
read	torchtitan/rl/observability/metrics/__init__.py	
read	torchtitan/rl/observability/metrics/console.py	
read	torchtitan/rl/observability/metrics/processor.py	
read	torchtitan/rl/observability/metrics/types.py	
read	torchtitan/rl/observability/rollout_recorder.py	
read	torchtitan/rl/observability/vllm.py	
read	torchtitan/rl/rollout/__init__.py	
read	torchtitan/rl/rollout/advantage.py	
read	torchtitan/rl/rollout/environment/__init__.py	
read	torchtitan/rl/rollout/environment/message.py	
read	torchtitan/rl/rollout/environment/token.py	

Event	Path or command	Hash
read	torchtitan/rl/rollout/rollouter.py	
read	torchtitan/rl/rollout/types.py	
read	torchtitan/rl/rubric.py	
read	torchtitan/rl/train.py	
read	torchtitan/rl/trainer.py	
read	torchtitan/rl/types.py	
read	torchtitan/tools/filesystem.py	
read	torchtitan/tools/garbage_collector.py	
read	torchtitan/tools/utils.py	
read	torchtitan/train.py	
read	torchtitan/trainer.py	
read	torchtitan/training_engine.py	
read	torchtitan_recipes/__init__.py	
read	torchtitan_recipes/graph_trainer/__init__.py	
read	torchtitan_recipes/graph_trainer/deepseek_v3.py	
read	torchtitan_recipes/graph_trainer/llama3.py	
read	torchtitan_recipes/graph_trainer/qwen3.py	
read	torchtitan_recipes/models/__init__.py	
read	torchtitan_recipes/models/deepseek_v3.py	
read	torchtitan_recipes/models/flux.py	
read	torchtitan_recipes/models/llama3.py	
read	torchtitan_recipes/models/muse_glimmer.py	
read	torchtitan_recipes/models/qwen3.py	
read	torchtitan_recipes/overrides/__init__.py	
read	torchtitan_recipes/overrides/fused_mla.py	
read	torchtitan_recipes/overrides/fused_swiglu.py	
read	torchtitan_recipes/overrides/helion_rope.py	
read	torchtitan_recipes/overrides/moe_token_dispatcher.py	
read	torchtitan_recipes/rl/__init__.py	
read	torchtitan_recipes/rl/alphabet_sort.py	
read	torchtitan_recipes/rl/dapo_math.py	
read	torchtitan_recipes/rl/search_r1.py	
read	torchtitan_recipes/rl/verifiers_dapo_math.py	
read	torchtitan_recipes/tests/__init__.py	
read	torchtitan_recipes/tests/datasets/__init__.py	
read	torchtitan_recipes/tests/datasets/synthetic_multimodal.py	
read	torchtitan_recipes/tests/graph_trainer/__init__.py	
read	torchtitan_recipes/tests/graph_trainer/b200.py	
read	torchtitan_recipes/tests/graph_trainer/deepseek_v3.py	
read	torchtitan_recipes/tests/graph_trainer/llama3.py	
read	torchtitan_recipes/tests/graph_trainer/muse_glimmer.py	
read	torchtitan_recipes/tests/graph_trainer/numerics.py	
read	torchtitan_recipes/tests/graph_trainer/precompile.py	
read	torchtitan_recipes/tests/graph_trainer/qwen3.py	
read	torchtitan_recipes/tests/models/__init__.py	
read	torchtitan_recipes/tests/models/deepseek_v3.py	
read	torchtitan_recipes/tests/models/deepseek_v4.py	
read	torchtitan_recipes/tests/models/flux.py	
read	torchtitan_recipes/tests/models/gpt_oss.py	
read	torchtitan_recipes/tests/models/kimi_k2_7.py	

Event	Path or command	Hash
read	torchtitan_recipes/tests/models/kimi_k3.py	
read	torchtitan_recipes/tests/models/llama3.py	
read	torchtitan_recipes/tests/models/multimodal.py	
read	torchtitan_recipes/tests/models/muse_glimmer.py	
read	torchtitan_recipes/tests/models/qwen3.py	
read	torchtitan_recipes/tests/models/qwen3_5.py	
read	torchtitan_recipes/tests/models/qwen3_6.py	
read	torchtitan_recipes/tests/models/qwen3_8.py	
read	torchtitan_recipes/tests/rl.py	
read	torchtitan_recipes/tests/suites/__init__.py	
read	torchtitan_recipes/tests/suites/b200.py	
read	torchtitan_recipes/tests/suites/features.py	
read	torchtitan_recipes/tests/suites/h100.py	
read	torchtitan_recipes/tests/suites/models.py	
read	torchtitan_recipes/tests/torchft/__init__.py	
read	torchtitan_recipes/tests/torchft/llama3.py	
read	torchtitan_recipes/tests/transformers_modeling_backend.py	
read	torchtitan_recipes/torchft/__init__.py	
read	torchtitan_recipes/torchft/llama3.py	
read	torchtitan.egg-info/PKG-INFO	
write	outputs/structured_logs/training.global_rank_0.20261008-044759-OK3XCR.j...	6f6f1dcf4c8b1b2c
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	grep -E step: +(1[0-9]*00)	
process	bash run_train.sh	
process	tee ../train.out	
process	sed s/^x1b\[0-9;\]*m//g	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python -u -m torchtitan.trai...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	pid 1362905	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	/var/tmp/kv-repro/026-pytorch-torchtitan/env/bin/python /var/tmp/kv-repro/0...	
process	python - ../train.out	

C Run history

Invocation 1, OK, 3 minutes 11 seconds, 66 output lines, started 2026-10-08T08:47:51

```
metric_hash c1b9d9c7cb3d2064950a4ea4626abf89ec4f05e3a13a14fbee5c13f6fe1e0bce
run_digest  8f634407a05fdd938ee8a8fd5c1ef6d28accd42d5e0518bf04d8a533eae6fa7
anchored    at run end
```

D Raw seal fields and hashes

Seal

```
data_hash          902ef28d56b69e5748e88bcc531d53db9a87875582631ae2eb9a8a81aae3c091
signed_message_hash 5f8ceaae1430e60efd4954238485b1627cf74de61daccfdc27c5f44c0db9f5ff
nonce             283ef67d5c7ac7ea57dc5a15b04105a0
signed_at         2026-10-08T08:51:05.194553+00:00
sealed_at         2026-10-08T08:51:05.256251726Z
checkout_bundle   9f8a20e45d71db5db150c421d2d32eccf42183e902e66d000b0e76348ec457a7
```

Signature (base64)

```
GRjyGMugJaLBzU9+RJQ6T0BH+x8ujyTnLzSK2eAvmZqVw4lpKhVLqL3soz1Ik3ANLF/SoS7XkZHWGdFP06cI9Byr
WBv0XsMwzBoKh3qkkwW1g3AD7Z48MMf6aaN/Q67lVmXorgriX7QIHYi4izM7MfmnCLZ258Z5dXk0SRPJ3tN+774E
5Z105SEcNrr7xMv8PfEdji6zC10nMPby++qq+fjeVVVea2s0EvoEVN5oz0Wn0mU0yve5joPhK1/mWPKYA/L4vk4C
UHN6wRdeYHhcNkbEhmxUKE7kiQt3mH1uCM8iaLayb0ItzJ6ZRLTqCKWYwHev4FDBAb3E6LsHoo7J0lAzhVBG1+bP
+AV7/stUidrH62urq/gS9rLoR9pw3r7NM3dPyDoVWjdxIBIC+co61yjnIUWYgm3JtG0URon//iJayZDlUvCl37tT
ekutoqddQCeyrh2RnlgecpDlneuyQMRU/1du5FzxhbDMcZ/ppZkydoUdCDCje3LtyvRWxUQIv0HFwfyI5Dz0bo0L
EaGau/44t8d8hilPQ7E3EyykaP3rjKw9mdctw0LpKtUHCQ0HgLBh6CRCsIvFava60sCXVglnud0SRX9gCcv5Hg6S
vH2p8j60D5xCSwrMvWsXLZcolUU4S/69zdDvFypi2Jcmb+huP+rT6h/lkw93oZadnVo=
```

Public key

```
-----BEGIN PUBLIC KEY-----
MIICIjANBgkqhkiG9w0BAQEFAAACAg8AMIICCgKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcPrOPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4LAt4vcgwP+iWscPc3HzpmjpsKbTmOemqgp6RB6GnnPpxYa5X
7qU95EfEisX/GfoS9ToxtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KICeAdbIbHN
av0JjdTJ//CM0WZSIbdzE6qvEbNFCqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHq+D
AhfTRuJNgWEkjwD5lb0Vhi4qlkweSOEMTKT3ye9Il5xEaJn4bku4eKqes7VSzt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlAQkJM6z2iITnyvDMgueJtFxQpC9
HNBPgPZ242NH8JgE6CNf7ToL9syasS67u7UMgIcNgeNHNcy9bLgpU5annzP9gN2l
6HEMldkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7lE3mbQRBpSLl/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/oLCHBCF1VIDCKDxoehdZ0WRRcgM9iK3I/0zZdaBSgdc8WuJRmb8LPP9v
8Jap/1PWjNcNku5cdw3gHLcCAwEAAQ==
-----END PUBLIC KEY-----
```

Run 1 outputs

```
stdout 281b44dcd9d3f9d259b0780b3418bff606ac44fcd734d8585df488ee1acc02b9
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics c1b9d9c7cb3d2064950a4ea4626abf89ec4f05e3a13a14fbee5c13f6fe1e0bce
digest  8f634407a05fdd938ee8a8fd5c1ef6d28accd42d5e0518bf04d8a533eae6fa7
```

kv_run.sh (unchanged)

```
pre 5290b881089a5bd48f9ab069a8dd70d5582a629a47b36686f63e570e2ddaddcd
post 5290b881089a5bd48f9ab069a8dd70d5582a629a47b36686f63e570e2ddaddcd
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,  
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},  
  author={Mamadou K. Keita and Christopher Homan},  
  year={2026},  
  eprint={2605.08586},  
  archivePrefix={arXiv},  
  primaryClass={cs.CR},  
  url={https://arxiv.org/abs/2605.08586},  
}
```